Q.1

In a 3×3 matrix, how many minors can be calculated?

A. 3
B. 9
C. **12**
D. 6

Q.2 Two matrices A and B are multiplied to get BA if

A. both are rectangular
B. both have same order
C. number of rows of A is equal to number of columns of B
D. **number of columns of A is equal to number of columns of B**

Q.3 For a geometric sequence, if 2nd term is equal to 4th term then value of common ratio is

A. 1 only
B. **+1 or -1**
C. 0
D. -1 only

Q.4

For a 3×3 matrix, swapping two rows will:

A. Leave the determinant unchanged.
B. Double the determinant.
C. **Square the determinant.**
D. Change the sign of the determinant.

Q.5 Cramer`s rule is not suitable for which type of problems?

A. Systems with 3 unknowns
B. **Large systems**
C. Systems with two unknowns
D. Small systems with 4 unknowns

Q.6

Find the minor of the element a_{11} in the matrix $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$.

JOIN Virtual University Whatsapp Group Click Here

A. 2
B. 4
C. **1**
D. 5

Q.7 What is the 31st term of the arithmetic sequence: 1, 4, 7, 10, …?

A. 90
B. 93**C. 91**
D. 92

Q.8 {\rm{The Cofactor of}}\,\,\,\,a_{ij} = \,\, - - - - .\,\,\,\,(Minor\,\,a_{ij})

A. ({ - 1})^{i + j}
B. None\,\,of\,\,these
C. ({ - 1})^{i j} \,
D. ({ - 1})^{i - j} \,

Q.9 What is the next term in the sequence 1, 2, 4, 8, 16, _____ ?

A. 36
B. **32**
C. 30
D. 24

Q.10 If the first term=a=1, common ratio=r=1/2; then which term of geometric sequence is 1/4?

A. 4
B. 6
C. **3**
D. 5

JOIN Virtual University Whatsapp Group Click Here

MTH100 MCQS
LMS HANDLING AND FILES 0336 0668195

Q.1 {\rm{The Cofactor of}}\,\,\,\,a_{ij} = \,\, - - - - .\,\,\,\,(Minor\,\,a_{ij})

A. ({ - 1})^{i + j}

B. ({ - 1})^{i j} \,

C. None\,\,of\,\,these

D. ({ - 1})^{i - j} \,

Q.2 What is the next term in the sequence 1, 4, 9, 16, 25, _____ ?

A. 40

B. 35**C. 36**

D. 30

Q.3 The explicit formula when a1 = 6 and d = 2 is _____.

A. an = 6 + 2(n – 1)

B. an = 6 + 2(n + 1)

C. an = 6 - 2(n – 1)D. an = 6 + 2n

Q.4 What is the term after the next term in the sequence 5, 10, 15, 20, _____?

A. 25

B. 35

C. 40

D. **30**

Q.5 A Square matrix with ones on the diagonal and zeros elsewhere is called a/an ..... matrix

A. identity

B. diagonal

C. Null

D. scalar

Q.6 Given a square matrix A, if there exists a matrix B such that AB = BA = I, then A is called the ..... of B

A. Additive inverse

B. **Multiplicative inverse**

C. Transpose

D. Adjoint

Q.7 The first term of the sequence whose A. 4

“nth term = 2n; n>2 “ is-----.

B. 6

C. 1

D. 2

JOIN Virtual University Whatsapp Group Click Here

Q.8 Two matrices of same order can be.....

A. subtracted but not added
B. added but not subtracted
C. multiplied
D. **added and subtracted**

Q.9 Cramer`s rule is not suitable for which type of problems?

A. Small systems with 4 unknowns
B. **Large systems**
C. Systems with two unknowns
D. Systems with 3 unknowns

Q.10 What is the common difference in the arithmetic sequence: 7, 11, 15, 19, ...?

A. 4
B. 5
C. 2
D. 3

JOIN Virtual University Whatsapp Group Click Here

MTH100 MCQS
LMS HANDLING AND FILES 0336 0668195

Q.1 Domain of function: f(x)=1/(Sqrt(x-1)) is all Real numbers ---------.

A. x<1
B. **x>1**
C. except x=1
D. except x=0

Q.2 If A = {1, 2, 3} and B = {2, 3, 4} then ……

A. Intersection of A and B has 3 elements
B. B is subset of A
C. A is subset of B
D. **Union of A and B has 4 elements**

Q.3 If f(x)=x and g(x)=1-x, then (f+g)(x) = ----------

A. 1-2x
B. 2x
C. **1**
D. 2x-1

Q.4 If f(x)=x and g(x)=1-x, then (f/g)(x) = ---------.

A. 1
B. 1-(1/x)
C. (1-x)/x
D. **x/(1-x)**

Q.5 Inverse of the function f(x)=3 doesn't exist due to which of the following reason?

A. It is many to one function
B. It is 1-1 function
C. It is Bijective function
D. It is Odd function

Q.6 If f(x)=x and g(x)=1-x, then (f-g)(x) = ----------

A. 1-2x
B. **2x-1**
C. 2xD. -1

Q.7 Which of these sets are equal {a,b,c}, {a,b,c,a},{a,b,b,c} and {a,c,b,c}?

A. 2nd and 3rd only
B. **All of these**
C. None of these
D. 1st and 4th only

Q.8 If\,f(x) = {x^2}\,and\,\,g(x) = \frac{1}{x},\,\,then\,\,fog(x) =

A. {1}{x}
B. **{1}{{{x^2}}}**
C. x
D. none

Q.9 Range of the function: f(x)=(1/x) is ---------.

A. All Real numbers except 1
B. **All Real numbers except 0**
C. All Real numbers
D. All Real numbers except -1

Q.10 2 is a

A. rational number
B. natural number
C. irrational number
D. integer

JOIN Virtual University Whatsapp Group Click Here

MTH100 MCQS
LMS HANDLING AND FILES 0336 0668195

Q.1 Every real number is -------.

A. a rational number
B. a positive integer
C. **a complex number**
D. a negative integer

Q.2 Which of the following is the real part of complex number: z = i?

A. 1
B. **0**
C. -1
D. None of these

Q.3 Domain of the function:f(x)=1/(1-x) is ---------.

A. All Real numbers
B. All Real numbers except -1
C. All Real numbers except 0
D. **All Real numbers except 1**

Q.4 \,{\mathop{\rm Re}\nolimits} al\,\,part\,\,of\,\,3ab\left({i + i^2 } \right)\,\,is\,\, - - - -

A. - 3abi
B. 3ab
C. **- 3ab**D. 3abi

Q.5 \begin{gathered} If A = \{ 1, 3, 5, 7, 9\} \hfill \ {\rm B} = \{ 2, 4, 6, 8, 10\} \hfill \ C = \{ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \hfill \ then what is A \cap B \cap C = ? \hfill \\ \end{gathered}

A. {}
B. \{ 2, 4, 6, 8, 10\}
C. \{ 1, 3, 5, 7, 9\}
D. \{ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}

Q.6 Cardinality of the Intersection of A={0,1,2} and B={1,2,3} is …….

A. 3
B. 1
C. 4
D. **2**

Q.7 Set of all the persons with horns on their heads is an example of…….

A. Empty set
B. Infinite Set
C. Finite Set with single element
D. Universal Set

Q.8 The\,\,value\,\,of\,\,z - \overline z \,\,is\,\, - - - - - .

A. 1
B. 0
C. **complex\,\,number**
D. rea{{ l}} \,\,number

Q.9 Cardinalities of sets {} and {{}} are……

A. Not equal
B. Equal

Q.10 Domain of function: f(x)=Sqrt(x-1) is all Real numbers ---------.

A. x>1
(note: Sqrt means square root of.)
B. x>1 and x=1
C. x<1
D. x<1 and x=1

JOIN Virtual University Whatsapp Group Click Here